Received: 28 October 2024 | Revised: 17 April 2025 | Accepted: 22 April 2025

DOI: 10.1002/alz.70277

Alzheimer's & Dementia®
THE JOURNAL OF THE ALZHEIMER'S ASSOCIATION

RESEARCH ARTICLE

Structural variation detection and association analysis of whole-genome-sequence data from 16,543 Alzheimer's disease sequencing project subjects

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Funding information
National Institute on Aging, Grant/Award Numbers: R01-AG074328, P30-AG072978, U54-AG052427, U24-AG041689, U01-AG032984, U01-AG058654, P30AG072979, U01-AG058654, U01-AG062602, R01-AG048927, P30-AG072978

Abstract

INTRODUCTION: The role of structural variations (SVs) in Alzheimer's disease (AD) remains understudied.

METHODS: We analyzed whole-genome sequencing data from the Alzheimer's Disease Sequencing Project ($N = 16,543$) and identified 400,234 (168,223 high-quality) SVs. Laboratory validation yielded a sensitivity of 82% (85% for high-quality).

RESULTS: We found a burden of singletons (odds ratio [OR] = 1.07, $p = 0.0017$) and homozygous deletions (OR = 1.14, $p < 0.0001$) in cases. On AD genes, we observed the ultra-rare SVs associated with the disease, including protein-altering SVs in *ABCA7*, *APP*, *PLCG2*, and *SORL1*. Twenty-one SVs are in linkage disequilibrium (LD) with known AD-risk variants, exemplified by a 5k deletion in LD ($R^2 = 0.99$) with rs143080277 in *NCK2*. We identified a rare deletion near *RNA5SP293* associated with AD (OR = 1.99, $p = 1.3 \times 10^{-5}$), which was replicated using an independent dataset.

DISCUSSION: This study highlights the pivotal role of SVs in AD genetics.

KEYWORDS
Alzheimer's disease, copy number variation, NCK2, RNA5SP293, structural variation

Highlights

- Observed a significant burden of singletons and homozygous deletions in Alzheimer's disease (AD) patients.
- Identified rare protein-altering structural variations (SVs) in ABCA7, APP, PLCG2, and SORL1.
- Established linkages between SVs and AD risk-associated single nucleotide variants (SNVs).
- Discovered a novel deletion near *RNA5SP293* linked to AD, replicated independently.
- Uncovered over-representation of SVs in neuronal function pathways.

1 | BACKGROUND

Alzheimer's disease (AD) is a neurodegenerative disease characterized by abnormal deposits of extracellular amyloid-β (Aβ) plaques and intracellular neurofibrillary tangles.[1] Typically, the accumulation of these neuropathological changes is accompanied by neuronal death, leading to various symptoms such as memory loss, apathy, difficulty swallowing, and walking.[2] Among individuals aged 65 years and older, AD has an incidence rate of 10.7% and is the fifth-leading cause of death.[2]

Genetic factors play a significant role in the etiology of AD, with the estimate of heritability ranging from 58% to 79%.[3] However, genetic risk factors identified in previous studies explain only a limited portion of heritability in AD. About 11% of early-onset AD, which is approximately 0.6% of all AD,[4] is caused by autosomal dominant mutations in *APP*, *PSEN1*, and *PSEN2*. Apolipoprotein E (*APOE)* genotype is the most prominent genetic risk factor for both early-onset and late-onset AD,[5,6] and it is estimated that approximately 40%–50% of individuals diagnosed with AD carry at least one copy of the *APOE* ε4 allele.[7] Particularly, *APOE* ε4 homozygosity represents a distinct genetic form of AD with near-full penetrance and an average age of symptom onset at 65.1.[8] Genome-wide association studies (GWASs) identified > 75 additional AD risk loci.[9–12] However, compared to *APOE* alleles, variants at those loci have a small effect size or are rare in the population, contributing little to the overall heritability. *APOE* alleles alone can achieve a similar or even better prediction accuracy of AD status than all other common single nucleotide variants (SNVs) combined.[13,14] When all common SNVs, including *APOE* alleles, are considered, they only account for 24%–33% of phenotypic variance,[15,16] which is much lower than the estimated heritability of AD (58%–79%)[3] and thus suggests a role for other genetic mechanisms.

Structural variants (SVs) are genomic alterations larger than 50 bp that include deletions, duplications, inversions, insertions, translocations, and complex combinations of these events. SVs contribute more to individual genetic variation in terms of total nucleotide content; thus, the difference in genomic sequences between two humans can increase from 0.1% with SNVs alone to 1.5% when SVs are considered.[17] Therefore, analyzing SVs has the potential to identify new genetic associations and account for the missing heritability in AD.

In AD, duplications in *APP* have been found to be the causal factor for autosomal dominant early-onset AD.[18–22] A deletion in exon 9 of *PSEN1* was identified in families with a form of early-onset AD characterized by spastic paraparesis and atypical plaques.[23,24] A low-copy repeat of 18 Kb in length within *CR1*, which creates an additional C3b/C4b-binding site, may account for the AD GWAS signals in the *CR1* region.[25,26] The 1 Mb region on 17q21.31 containing *MAPT* has two major haplotypes, H1 and H2, which are characterized by a ~900 Kb inversion tagged by a 238 bp deletion between exons 9 and 10 of *MAPT*.[27] The H1 and H2 haplotypes are associated with a range of neurodegenerative diseases, including AD.[27–29] Additionally, copy number variants in *AMY1*, which are correlated with salivary amylase protein level and digestion of starchy food, are associated with AD. Individuals with high copy numbers ($\geq$10) of *AMY1* have a significantly lower risk of developing AD.[30] These examples show that the identification and analysis of SVs in AD genetics hold great potential for uncovering new genetic associations.

To discover SV variants contributing to AD risk, we evaluated SVs detected in paired-end (PE) short read data from whole-genome sequence (WGS) of 16,905 DNA samples from 16,543 subjects from the Alzheimer's Disease Sequencing Project (ADSP). The dataset comprises a mixture of 100 bp PE reads (from Illumina HiSeq2000/2500) and 150 bp PE reads (from Illumina HiSeq X/Ten and Nova Seq). We detected 400,234 SVs and found rare SVs in known AD genes, including *SORL1*, *ABCA7* and *APP*, as well as SVs in linkage disequilibrium (LD) with AD GWAS signals. Moreover, we found an increased burden of deletions (particularly, singleton and homozygous events) in AD and identified risk SVs through association analysis.

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2 | METHODS

2.1 | Study subjects

ADSP[31] is a collaborative project aiming at identifying new variants, genes, and therapeutic targets in AD. In the R3 release of ADSP, 16,905 DNA samples from 16,543 subjects were collected across 24 cohorts, and WGS was performed by Illumina HiSeqX, HiSeq2000, HiSeq2500, and NovaSeq platforms. Demographic characteristics for 16,543 subjects were displayed in Table 1. The ancestry of each individual was inferred using GRAF-pop.[32] The samples came from diverse ancestries with 10,184 Europeans, 3574 African Americans, 2643 Latin Americans, 57 East Asians, and 85 of other ancestries. There are 6525 AD cases, 6855 controls, and 3163 subjects with unknown status in this study.

TABLE 1 Characteristics of study subjects.

Parameter	AD (*N* = 6525)	Control (*N* = 6855)	Unknown AD status (*N* = 3163)
Age (SD)	74.65 (10.43)	77.39 (7.97)	72.24 (9.15)
Sex (%)			
Female	3925 (60.15%)	4582 (66.84%)	1503 (47.52%)
Male	2600 (39.85%)	2273 (33.16%)	1660 (52.48%)
APOE status[a] (%)			
ε2ε2	14 (0.21%)	35 (0.52%)	20 (0.65%)
ε2ε3	286 (4.39%)	659 (9.74%)	296 (9.57%)
ε2ε4	145 (2.23%)	141 (2.08%)	56 (1.81%)
ε3ε3	2726 (41.84%)	3914 (57.83%)	1855 (59.95%)
ε3ε4	2774 (42.58%)	1855 (27.41%)	761 (24.60%)
ε4ε4	570 (8.75%)	164 (2.42%)	106 (3.43%)
Ancestry[b] (%)			
European	4289 (65.73%)	3160 (46.10%)	2735 (86.47%)
African American	1428 (21.89%)	2021 (29.48%)	125 (3.95%)
Latin American	765 (11.72%)	1639 (23.91%)	239 (7.56%)
East Asian	18 (0.28%)	9 (0.13%)	30 (0.95%)
Other	25 (0.38%)	26 (0.38%)	34 (1.07%)
Ethnicity[c] (%)			
Non-Hispanic	5417 (84.14%)	4845 (70.92%)	1034 (80.66%)
Hispanic	1021 (15.86%)	1987 (29.08%)	248 (19.34%)

Note: After removing 362 duplicates, 16543 samples were included in the table.
Abbreviations: AD, Alzheimer's disease; Age, age at onset for individuals with AD or age at last exam for non-AD subjects; APOE, apolipoprotein E; SD, standard deviation.
[a]*APOE* status is based on rs429358 and rs7412 observed from whole genome sequencing data.
[b]Ancestry is inferred using GRAF-pop.[32]
[c]Ethnicity is self-reported.

RESEARCH IN CONTEXT

1. **Systematic review**: We reviewed peer-reviewed (PubMed) and preprint articles analyzing whole-genome sequencing (WGS) data to identify structural variants (SVs) linked to diseases. Our analysis focused on Alzheimer's Disease Sequencing Project WGS data to investigate SVs in Alzheimer's disease (AD).
2. **Interpretation**: This is the first large-scale study of SVs in AD genomes using WGS data. We found a significant burden of singletons and homozygous deletions in AD patients. Among AD-associated genes, we identified ultra-rare, protein-altering SVs in *ABCA7*, *APP*, *PLCG2*, and *SORL1*. Additionally, 21 SVs were in linkage disequilibrium (LD) with known AD-risk variants, including a 5k deletion in perfect LD with rs143080277 in *NCK2*. A novel rare deletion near *RNA5SP293*, associated with AD, was also replicated in an independent dataset.
3. **Future directions**: Our study provides new insights into the role of SVs in AD. Further functional validation of these variants and replication in larger WGS datasets will be critical.

After removing duplicates and subjects without AD diagnosis, 13,371 samples were kept for analysis. Then, 463 outlier subjects, with too many (> median + 4 * mean absolute deviation) SV calls or too few (< median − 4 * mean absolute deviation) high-quality SV calls, were removed (Figure S1). There were 12,908 samples (6328 cases and 6580 controls) remaining for association analysis (Figure S2). Family relationships among these samples were checked by KING (v2.3.2),[33] identifying 4926 pairwise relationships (second-degree or closer) involving 2363 samples. Outliers are more likely to be of smaller insert size (327.62 ± 40.82) compared to the samples that were kept for further analysis (401.00 ± 41.33) (Wilcoxon rank-sum test, $p = 2.53 \times 10^{-170}$) (Figure S3). Outliers have a higher average coverage (40.63 ± 14.84 vs. 40.37 ± 5.90, Wilcoxon rank-sum test, $p = 1.75 \times 10^{-8}$), while they also exhibit a peak with low coverage (Figure S3). In ADSP R4, an additional 19,456 study subjects were released. We selected 15,528 subjects (5051 cases and 10,477 controls) with diagnosis of AD for replication of significant SVs from the discovery phase. Sample characteristics for replication samples were displayed in Table S1.

2.2 | SV calling

Figure S4 illustrates the SV calling pipeline. In the discovery phase, SVs were called by Manta[34] (v1.6.0) and Smoove[35] (v0.2.5) with default parameters for each sample. Calls from Manta and Smoove were merged by Svimmer[36] to generate a union of two call sets for a sample.

Unresolved non-reference "breakends" (BNDs) and SVs > 10 Mb were filtered. Then, all individual sample VCF files were merged together by Svimmer as input to GraphTyper2 (v2.7.3)[36] for joint genotyping. SV calls after joint-genotyping are comparable across the samples; therefore, they can be used directly in genome-wide association analysis.[36] In the replication phase, significant SVs from the association analysis of discovery data were selected and joint-genotyped by GraphTyper2. The pipeline is available on https://github.com/whtop/SV-ADSP-Pipeline.

2.3 | High-quality SVs

A subset of SV calls was defined as high-quality calls. The criteria for high-quality SVs can be found in GraphTyper2 study[36]: For deletion, QD (QUAL divided by non-reference sequence depth) > 12 & (ABHet [allele balance for heterozygous calls (read count of call2/(call1 + call2]) where the called genotype is call1/call2, -1 if no heterozygous calls.) > 0.30 | ABHet < 0) & (AC / NUM_MERGED_SVS [number of SVs merged]) < 25 & PASS_AC (number of alternate alleles in called genotyped that have "FT" field as "PASS") > 0 & PASS_ratio (ratio of genotype calls that have "FT" field as "PASS") > 0.1; For duplication, QD > 5 & PASS_AC > 0 & (AC / NUM_MERGED_SVS) < 25; For insertion, PASS_AC > 0 & (AC/NUM_MERGED_SVS) < 25 & PASS_ratio > 0.1 & (ABHet > 0.25 | ABHet < 0) & MaxAAS (maximum alternative allele support per alternative allele) > 4; For inversions: PASS_AC > 0 & (AC/NUM_MERGED_SVS) < 25 & PASS_ratio > 0.1 & (ABHet > 0.25 | ABHet < 0) & MaxAAS > 4.

2.4 | SV selection by algorithmic models

GraphTyper2 generates calls for each SV by algorithmic models, that is, breakpoint (including breakpoint1 and breakpoint2), coverage, and aggregated models.[36] Note that an SV call can be annotated by multiple models, so there will be duplicated records in VCF if an SV call has more than one algorithmic model. Deletions and duplications were called by aggregated, coverage, and breakpoint models (Table S2). Insertions and inversions were called by aggregated and breakpoint models (Table S2). The aggregated model has the highest recall of the other two models.[36] To select a single SV call per structural variant, we first prioritize high-quality calls. If multiple models generate calls for the same SV, the selection follows this order of priority: aggregated calls, breakpoint calls, and then coverage calls. If no high-quality calls are available, we still select a call based on the same order: aggregated calls first, followed by breakpoint calls, and finally coverage calls (Table S2).

2.5 | Problematic regions

There are regions in the human genome that tend to have anomalous, or high signal, in WGS experiments.[37] SVs that reside in those regions can be unreliable and should be reported. Specifically, we compiled problematic regions in the genome from the following sources: (1) the ENCODE blacklist: a comprehensive set of regions that could result in erroneous signal[38]; (2) the 1000 Genome masks: regions of the genome that are more or less accessible to next generation sequencing methods using short reads; (3) the set of assembly gaps defined by UCSC; (4) the set of segmental duplications defined by UCUC; (5) the low-complexity regions, satellite sequences and simple repeats defined by RepeatMasker.[39]

2.6 | High-confident SVs

For any SVs reported on AD risk/protective genes and from association, Samplot[40] was used to check their alignment supports of read depth and/or split reads if SV types are deletions, duplications, and inversions. For insertions, which cannot be inspected using Samplot, we kept insertions that are high-quality and not in the problematic regions.

2.7 | SV annotation

SVs were annotated using Variant Effect Predictor (VEP, Ensembl V 107)[41] and annotSV (V 3.1.1).[42] SVs that were annotated (by VEP) to be able to cause transcript ablation/amplification, stop gain, start/stop lost, frameshift, inframe deletion/insertion, missense mutation, and affecting splice acceptor/donor were classified as protein-altering variants. The impact of SVs is also evaluated by the annotSV ranking score, which is an adaptation of the work provided by the joint consensus recommendation of the American College of Medical Genetics and Genomics (ACMG) and ClinGen.[43]

2.8 | SV validation

SVs from the 1000 Genomes Project phase III[44] and gnomAD[45] were downloaded from the dbVar database[46] with study accession ID estd219 and nstd166. On chromosomes 1–22, there are 66,505 and 292,307 SVs from the 1000 Genomes Project and gnomAD, respectively. For deletions/duplications/inversions, calls with at least 50% reciprocal overlapping were considered as replicated. For insertions, we searched for calls with breakpoints within 500 bp. Then, we estimated the sensitivity of GraphTyper2 (Table S3) by synthetic mutations (i.e., "spiking-in" SVs) generated from three samples by Malamon et al.[47]

For polymerase chain reaction (PCR) validation, we assessed a total of 95 SVs, including 41 SVs near AD related genes, 32 randomly selected SVs, and 22 SVs that can cause loss of function for the gene (Table S4). At the SV site level, experimental validation of at least one DNA sample served to authenticate the existence of the SV within the genome. Then, the sensitivity was determined by dividing the number of confirmed SVs by the total number of SVs subjected to validation, which is 82% (78 out of 95). For each SV, we typically selected two to three samples for validation, resulting in a total of 276 samples across

95 SVs. At the individual genotype level, the genotype determined by GraphTyper2 for a specific sample was compared against experimental validation. Given that there could be calls from aggregated, breakpoint, and coverage models for each sample (534 genotype calls for 276 samples), we computed accuracy, precision, recall, and F1 scores for each model separately (Table S5). Subsequently, overall accuracy, precision, recall, and F1 scores were computed based on SV calls from all models (also detailed in Table S5).

The sequences surrounding the variants were extracted and used to design PCR primers. For deletions under 1100 bp, primers were designed outside of the breakpoints to amplify across the deletion sequence. For deletions where the reference allele was too large to be amplified by PCR, a double PCR approach was used. For duplication variants, since most duplications occur in a head-to-tail orientation, PCR primers were designed to amplify a product in this orientation. A forward direction primer was designed at the 3′ end of the duplicated sequence, and a reverse primer was designed at the 5′ end of the duplicated sequence. The detailed protocol can be found in the Supplementary Materials.

2.9 | SVs on AD risk loci and AD genes

We first searched for SVs that are in LD with AD associated loci from three GWASs.[10–12] There are 123 unique variants that reached genome-wide significance from three studies. After excluding nine variants that were not found in the WGS data, we searched for SVs that are in LD ($R^2 > 0.2$) with the rest of 114 variants. For these SVs and SNVs, we conducted an association analysis comparing AD cases and controls ($N = 12{,}908$) using a generalized linear mixed model (–fastGWA-mlm-binary) implemented in genome-wide complex trait analysis (GCTA).[48] Subject relatedness was adjusted using a sparse genetic relationship matrix derived from SNV data, applying a cutoff value of 0.05. The model was adjusted for PCs 1-5, age, sex, sequencing centers, sequencing platforms, and PCR status. Age was defined as the age at disease onset for cases and the age at the last visit for controls. PCR status was included because PCR amplification can introduce systematic biases in base composition,[49] potentially leading to inaccurate SV calls. We reported allele frequency (AF), odds ratio, and *p*-value from association analysis for both SNVs and SVs in Table 2. Additionally, Table 2 reports functional annotations for both SNVs and SVs, including nearest genes, distance from candidate cis-Regulatory Elements (cCREs) in ENCODE,[50] density of regulatory annotations from the ReMap Atlas.[51] The ReMap Altas consists of a large-scale integrative analysis of all public ChIP-seq data for transcriptional regulators.

Then, we investigated SVs on known AD genes. A list of 20 expert-curated AD risk/protective causal genes was downloaded from: https://adsp.niagads.org/index.php/gvc-top-hits-list. These genes were identified by a review of the literature, pathway analysis, and by integration of genetic studies with myeloid genomics. All deletions, duplications, and inversions with a missing rate less than 0.5 that overlap with these genes were inspected by Samplot. Deletions and duplications that passed inspection are listed in Table 3 and Table S6. High-quality insertions on 20 AD genes with a missing rate less than 0.5 and not located on problematic regions were reported in Table S7. For SVs with a minor allele count ≥ 5, association analysis was performed using the same generalized linear mixed model (–fastGWA-mlm-binary) implemented in GCTA[48] adjusting for covariates and subject relatedness. Association of ultra-rare SVs (minor allele count < 5) on 20 AD genes was evaluated using the SKAT-O test from the R package SKAT.[52] Additionally, we included a full list of SVs on other possible AD genes identified from three GWASs[10–12] in Table S8.

2.10 | Overall SV burden in AD

Overall SV burden between AD cases and controls was compared. SV burden was measured by the difference in the number of high-quality SVs in cases and controls. Related samples (second degree or closer) were excluded from the analysis, resulting in a final dataset of 12,089 samples, comprising 5858 cases and 6231 controls. Logistic regression models were implemented using the glm function in R (v4.2.1)[53] and adjusted for covariates including PCs 1–5, age, sex, sequencing center, sequencing platform, and PCR status. SV count used in the logistic regression model was standardized by subtracting mean then divided by standard deviation. One-sided empirical *p*-values (assuming increased SV burden in cases) were calculated based on 10,000 permutations. Particularly, we evaluated the burden of singletons and homozygous SVs in AD compared to controls.

2.11 | Association and functional analyses

In the discovery phase, 136,092 SVs in 12,908 subjects (6328 cases and 6580 controls) with a missing rate < 0.5 and minor allele count ≥ 5 were evaluated using generalized mixed linear model (–fastGWA-mlm-binary) implemented in GCTA.[48] Age, sex, sample PCR status, sequencing platforms, sequencing centers, and PCs 1–5 calculated from common SNVs were included as covariates. The age of cases was determined by the age at disease onset. The age of controls was determined by the age at the last exam. Sparse genetic relationship matrix was generated using SNVs as well with a cutoff of 0.05. High-confident deletions, duplications, and inversions were selected by Samplot and experimentally validated by PCR. For insertions, only high-confident ones that are high-quality and not on the problematic regions were reported. A total of 2,234 high-quality SVs with nominal significance ($p < 0.05$) were annotated to their nearest genes using the annotatePeak function from ChIPseeker.[54,55] Enrichment analysis for the 1945 genes located on or near 2234 SVs was performed using clusterProfiler.[56] In the replication phase, we evaluated significant SVs (FDR < 0.2) from the discovery phase using an additional 15,528 subjects from ADSP R4 (Table S1). The same model implemented by GCTA and the same covariates as the discovery phase were used in the replication analysis. We also performed a pooled analysis that combined samples from the discovery and replication phases. Association analysis was also conducted separately for each ancestry group (Europeans,

TABLE 2 High-confident SVs in linkage disequilibrium with AD GWAS signals

SNV									SV							
SNV	Position	Original P	Distance to cCRE	ReMap density	Gene	AF	OR (95% CI)	p^a	SV	Distance to cCRE	ReMap density	Gene	AF	OR (95% CI)	p^a	R^2
rs115186657	chr2:105618971	1.30E-08[b]	376	1	ENSG00000287308 intron	0.004	1.36 (0.87–2.13)	0.17	chr2:105731359-105736864:DEL[*]	0	130	Intergenic (near NCK2)	0.005	1.33 (0.9–1.97)	0.16	0.65
rs143080277	chr2:105749599	2.10E-13[c]	0	23	NCK2 intron	0.005	1.32 (0.89–1.95)	0.17								0.99
rs139643391	chr2:202878716	1.10E-08[c]	2932	1	WDR12 exon	0.095	1.06 (0.97–1.16)	0.20	chr2:203034369-203039560:DEL	0	156	NBEAL1 intron	0.099	1.06 (0.97–1.17)	0.18	0.94
rs5011436	chr7:12229132	2.70E-09[b]	2005	2	TMEM106B intron	0.500	1 (0.94–1.05)	0.88	chr7:12242077-12242399:DEL	1854	0	TMEM106B exon	0.541	1.01 (0.96–1.07)	0.67	0.92
rs13237518	chr7:12229967	4.90E-11[c]	2840	8	TMEM106B intron	0.489	1 (0.95–1.05)	0.95								0.9
rs1160871	chr7:28129126	9.80E-09[c]	1127	4	JAZ1 intron	0.377	0.95 (0.9–1)	0.07	chr7:28174681-28175045:DEL	1267	8	JAZ1 intron	0.388	1.05 (1–1.11)	0.07	0.41
rs7908662	chr10:122413396	2.60E-09[c]	3710	5	PLEKHA1 intron	0.444	0.97 (0.92–1.03)	0.31	chr10:122457302-122457747:DEL	232	2	ARMS2 exon	0.227	0.98 (0.92–1.05)	0.58	0.24
rs6489896	chr12:113281983	1.80E-09[c]	2883	1	TPCN1 intron	0.101	1.02 (0.93–1.12)	0.65	chr12:113245316-113245625:DEL	28	46	TPCN1 intron	0.136	1.03 (0.94–1.13)	0.50	0.73
rs10131280	chr14:106665591	4.30E-10[c]	928	6	Intergenic (near IGHV3-65)	0.157	0.96 (0.89–1.03)	0.24	chr14:106774952-106775261:DEL	0	31	IGHV3-71 (Pseudogene) exon	0.137	0.97 (0.89–1.06)	0.51	0.26
rs199515	chr17:46779275	9.30E-13[c]	0	18	WNT3 intron	0.185	0.94 (0.87–1)	0.06	chr17:46009357-46009595:DEL	485	8	MAPT intron	0.167	0.93 (0.86–1)	0.04	0.67
									chr17:46099028-46099351:DEL	1569	5	KANSL1 intron	0.203	0.92 (0.86–0.99)	0.02	0.61
									chr17:46146541-46146855:DEL	0	10	KANSL1 intron	0.181	0.92 (0.86–0.99)	0.03	0.62
									chr17:46205463-46208952:DUP	2703	12	KANSL1 intron	0.061	0.94 (0.83–1.06)	0.31	0.35
									chr17:46237501-46238225:DEL	11818	1	Intergenic (near MAPK8IP1P1)	0.187	0.96 (0.89–1.05)	0.38	0.54
									chr17:46277789-46282210:DEL	2331	212	ARL17B intron	0.113	0.91 (0.83–1)	0.05	0.59
									chr17:46135409-46292152:DUP	0	1775	Partial duplication of KANSL1 and ARL17B	0.324	0.98 (0.93–1.04)	0.48	0.42
rs10947943	chr6:41036354	1.10E-09[c]	0	48	UNC5CL intron	0.107	0.94 (0.87–1.03)	0.19	chr6:40959079-40959079:INS	668	17	Intergenic (near UNC5CL)	0.210	0.95 (0.88–1.02)	0.16	0.23
rs3740688	chr11:47358789	5.40E-13[d], 8.78E-9[b]	266	38	SPI1 exon	0.394	0.97 (0.92–1.03)	0.33	chr11:47775210-47775210:INS	1154	1	Intergenic (near NUP160)	0.262	0.97 (0.89–1.06)	0.51	0.28
rs6489896	chr12:113281983	1.80E-09[c]	2883	1	TPCN1 intron	0.101	1.02 (0.93–1.12)	0.65	chr12:113286417-113286417:INS	424	19	TPCN1 intron	0.078	1.06 (0.95–1.18)	0.31	0.69

(Continues)

TABLE 2 (Continued)

SNV									SV							
SNV	Position	Original P	Distance to cCRE	ReMap density	Gene	AF	OR (95% CI)	p^a	SV	Distance to cCRE	ReMap density	Gene	AF	OR (95% CI)	p^a	R^2
rs7146179	chr14:52832135	6.99E-1[b]	1230	22	ENSG00000285664 intron	0.177	1.05 (0.98–1.12)	0.18	chr14:52832930-52832930:INS	435	6	ENSG00000285664 intron	0.766	1.01 (0.93–1.11)	0.75	0.8
rs28394864	chr17:49373413	4.90E-10[b]	667	1	ENSG00000262039 intron	0.405	0.98 (0.93–1.03)	0.41	chr17:49320942-49320942:INS	3241	2	ZNF652 intron	0.277	0.91 (0.82–1)	0.05	0.41
rs138190086	chr17:63460787	7.50E-09[d]	32	257	Intergenic (near ENSG00000226597)	0.013	1.2 (0.95–1.52)	0.13	chr17:63204093-63204093:INS	2141	4	TANC2 intron	0.067	1.01 (0.89–1.16)	0.83	0.23
rs2154482	chr21:26148613	7.66E-10[b]	168	60	APP intron	0.471	1.01 (0.95–1.06)	0.84	chr21:26136136-26136136:INS	303	1	APP intron	0.157	1.08 (0.97–1.2)	0.16	0.4

Note: Distance to cCRE (cis-Regulatory Elements): distance from candidate cCREs in ENCODE.[50] ReMap Density: Density of annotations from ReMap Atlas[51] of regulatory regions, which consists of a large-scale integrative analysis of all public ChIP-seq data for transcriptional regulators.

Abbreviations: AD, Alzheimer's disease; AF, allele frequency; CI, confidence interval; GCTA, genome-wide complex trait analysis; GWAS, genome-wide association studies; OR, odds ratio; SNV, single nucleotide variant; SV, structural variant;

[a]p-values from association analysis by –fast-mlm-binary using GCTA.

[b]Wightman et al., 2021.

[c]Bellenguez et al., 2022.

[d]Kungle et al., 2019.

*SVs that have been experimentally validated.

African Americans, and Latin Americans), followed by meta-analysis across ancestries using METAL (release 20200505).[57] Given that age of onset is a discriminating factor in AD genetics, we further analyzed late-onset cases (≥ 65) versus controls, as well as early-onset cases (< 65) versus controls.

In addition to binary AD diagnosis, we also assessed SV associations with cognitive scores, fluid biomarkers, and neuropathological measurements using a mixed linear model (–fastGWA-mlm) implemented in GCTA,[58] adjusting for the same set of covariates and subject relatedness. These phenotypes were harmonized by the ADSP Phenotype Harmonization Consortium.[59] Cognitive scores include memory score ($N_{discovery}$ = 6,413, $N_{replication}$ = 2,688), executive function score ($N_{discovery}$ = 5,762, $N_{replication}$ = 2,251), language score ($N_{discovery}$ = 6,130, $N_{replication}$ = 2,552), and visuospatial score ($N_{discovery}$ = 1,126, $N_{replication}$ = 359).[59] Fluid biomarkers include cerebrospinal fluid (CSF) amyloid beta ($N_{discovery}$ = 1,110, $N_{replication}$ = 251), tau ($N_{discovery}$ = 1,086, $N_{replication}$ = 244), and P-tau ($N_{discovery}$ = 1,087, $N_{replication}$ = 236). Neuropathological measurements include Thal amyloid phases ($N_{discovery}$ = 543, $N_{replication}$ = 445), CERAD amyloid scores ($N_{discovery}$ = 2,361, $N_{replication}$ = 842), amyloid presence (dichotomous, $N_{discovery}$ = 2,361, $N_{replication}$ = 843), BRAAK tau phases ($N_{discovery}$ = 2,357, $N_{replication}$ = 828), ADNC severity scores ($N_{discovery}$ = 540, $N_{replication}$ = 444), and NIA-REAGAN criteria for AD ($N_{discovery}$ = 1,060, $N_{replication}$ = 358).

3 | RESULTS

3.1 | SV discovery and characteristics

The SV discovery pipeline, including the Manta,[34] Smoove,[35] Svimmer,[36] and GraphTyper2[36] SV callers (Section 2), was applied to ADSP[31] R3 release (NG00067.v7) WGS data (*N* = 16,543; Table 1). We observed 400,234 SVs (231,385 deletions, 45,839 duplications, 119,648 insertions, and 3362 inversions) of which 168,223 (98,805 deletions, 24,602 duplications, 44,130 insertions, and 506 inversions) were classified as high-quality calls (Table S2, Section 2). Notably, genotype calls for deletions exhibited superior quality with a higher pass ratio (the percentage of individuals passed quality control for the SV call) compared to duplications, insertions, and inversions (Figure S5). This observation highlights the higher quality of deletion detection on WGS over other SV types using available callers.

On average, each individual had 14,607 (3875 high-quality) deletions, 764 (288 high-quality) duplications, 6980 (2504 high-quality) insertions, and 19 (3 high-quality) inversions. Individuals of African ancestry had more SV calls compared to individuals of other ancestries (Figure 1A), possibly because the human reference genome is biased toward European ancestry or a higher level of genetic diversity in Africans.[60–62] Similar to SNVs, the first two principal components of common SVs distinguished samples from different ancestral backgrounds (Figure 1B). However, the third principal component of SVs is associated with read length and sequencing platforms (Figure S6),

TABLE 3 Ultra-rare SVs on AD genes

SV	Size	AC (case) [AgeOnset, Sex, Eth]	AC (control) [Age, Sex, Eth]	Gene	Type	Protein-altering[a]
chr19:1050368-1050973[b]	605	4 [74, F, AA] [69, M, AA] [81, F, AA] [81, F, AA]	0	ABCA7	DEL	Yes
chr16:81775821-81829769[b]	53,948	4 [75, M, E] [56, F, E] [-, M, E] [-, M, E]	0	PLCG2	DEL	Yes
chr19:1052156-1060559	8403	3 [72, F, AA][d] [72, F, AA]	1 [70, F, L]	ABCA7	DUP	Yes
chr16:81860089-81940500	80,411	3 [89, F, E] [56, F, E] [-, F, E]	0	PLCG2	DEL	Yes
chr2:127094683-127094740[b]	57	2 [86, M, E] [87, F, E]	1 [84, F, E]	BIN1	DEL	No
chr7:100296733-100385675	88,942	2 [85, F, E] [56, F, E]	0	PILRA	DEL	Yes
chr16:81907252-81907401	149	2 [67, F, E] [61, F, E]	0	PLCG2	DEL	No
chr14:92611452-92611515	63	2 [90, M, E] [61, F, E]	0	RIN3	DEL	No
chr19:1043504-1053484	9980	1 [78, M, E]	0	ABCA7	DEL	Yes
chr19:1054326-1061615[b]	7289	1 [67, M, E]	0	ABCA7	DUP	Yes
chr15:58720431-58721649	1218	1 [79, F, AA]	0	ADAM10	DEL	No
chr21:25815144-26232105[e]	416,961	1 [52, M, E]	0	APP	DUP	Yes
chr21:25958556-25971275	12,719	1 [-, F, E]	0	APP	DEL	No
chr21:26163874-26163976	102	1 [70, M, E]	0	APP	DUP	No
chr2:127102503-127104954	2451	1 [70, M, AA]	0	BIN1	DEL	No
chr10:113725274-113726288	1014	1 [75, F, E]	0	CASP7	DEL	Yes
chr11:60138757-60178011	39,254	1 [65, F, AA]	1 [82, F, E]	MS4A6A	DEL	Yes
chr11:60179765-60450406[b]	270,641	1 [-, F, O]	0	MS4A6A	DUP	Yes
chr11:86050643-86054032	3389	1 [70, F, AA]	0	PICALM	DEL	No
chr16:81734058-81749307[c]	15,249	1 [-, F, E]	0	PLCG2	DEL	Yes
chr16:81749081-81749132	51	1 [75, M, AA]	0	PLCG2	DEL	No
chr16:81755550-81764402[c]	8852	1 [84, F, E]	0	PLCG2	DEL	Yes
chr16:81772599-81777744	5145	1 [69, F, E]	0	PLCG2	DEL	Yes
chr16:81792449-81792587	138	1 [68, F, E]	0	PLCG2	DEL	No
chr16:81798030-81802305[b]	4275	1 [72, M, E]	1 [89, F, AA]	PLCG2	DEL	Yes
chr16:81822810-81822862[b]	52	1 [83, M, E]	0	PLCG2	DEL	No
chr16:81868226-81868350	124	1 [66, F, AA]	1 [70, M, AA]	PLCG2	DEL	No
chr14:92369331-92644481	275,150	1 [85, M, AA]	0	RIN3	DUP	Yes

(Continues)

TABLE 3 (Continued)

SV	Size	AC (case) [AgeOnset, Sex, Eth]	AC (control) [Age, Sex, Eth]	Gene	Type	Protein-altering[a]
chr14:92531510-92573409	41,899	1 [-, F, E]	0	RIN3	DUP	Yes
chr11:121303351-121495669	192,318	1 [61, F, AA]	0	SORL1	DUP	Yes
chr11:121490959-121499413	8454	1 [70, F, E]	0	SORL1	DEL	Yes
chr19:1051380-1051420	40	0	1 [81, M, AA]	ABCA7	DEL	No
chr15:58621731-58622007	276	0	1 [64, F, L]	ADAM10	DUP	No
chr19:44909364-44909819	455	0	1 [80, M, AA]	APOE	DUP	Yes
chr21:26013065-26013159	94	0	1 [69, F, AA]	APP	DEL	No
chr2:127064222-127064288	66	0	1 [89, F, AA]	BIN1	DEL	No
chr2:127092466-127092526	60	0	1 [68, M, L]	BIN1	DEL	No
chr2:127094016-127102983	8967	0	1 [80, F, E]	BIN1	DEL	No
chr1:207604022-207605343	1321	0	1 [-, F, L]	CR1	DEL	No
chr7:100377717-100378235	518	0	1 [64, M, AA]	PILRA	DEL	No
chr16:81746327-81746435[b]	108	0	1 [85, F, E]	PLCG2	DEL	No
chr16:81885235-81893072	7837	0	1 [73, F, AA]	PLCG2	DEL	Yes
chr14:73215181-73313643	98,462	0	1 [84, F, E]	PSEN1	DEL	Yes
chr14:92535796-92535871[b]	75	0	1 [85, F, E]	RIN3	DEL	No
chr14:92605335-92607867[b]	2532	0	1 [77, F, E]	RIN3	DEL	No

Note:"-" represents missing age. AD risk/protective genes are selected by ADSP Gene Verification Committee (https://adsp.niagads.org/gvc-top-hits-list/).
Abbreviations: A, Asian; AA, African American; AD, Alzheimer's disease; E, European, F, female; L, Latin American; M, male; O, other; PCR, polymerase chain reaction; SV, structural variant.
[a]SV overlaps with gene exons.
[b]SVs that are experimentally validated.
[c]SVs with read depth and split reads support but no PCR product for flanking primers.
[d]Represents homozygous even.
[e]Two additional family individuals had the duplication but failed quality control due to less confident genotypes. Nevertheless, alignment evidence strongly supports the presence of duplication. Their onset ages are 49 and 53.

indicating batch effect is an important confounding factor to consider when performing SV analysis.

Comparable to the AF distribution of SNVs, most SVs are extremely rare. Among 400,234 SVs, 94,923 (24%) are singletons, and 232,295 (58%) are rare with AF < 1% (Figure S7). When considering the 168,223 high-quality SVs, 67,595 (40%) are singletons, and 140,164 (83%) are rare with AF < 1%. Figure 1C shows that the AF distribution of deletions is more similar to the AF distribution of SNVs compared to other SV types. Analysis of the size of the SVs revealed two peaks centered around 300 bp and 6000 bp (Figure 1D and E), suggesting the possibility that many SVs are introduced by transposons, particularly Alus (~300 bp) and L1s (~6000 bp).[45]

Functional annotation analysis performed using AnnotSV[42] showed that rare SVs are more likely to be deleterious than common SVs (Wilcoxon rank sum $p < 0.0001$) (Figure 2A). This finding was confirmed using annotation from VEP,[41] which shows that protein-altering SVs tend to be rare (odds ratio [OR] = 4.71, chi-squared $p < 0.0001$, Figure 2B). Additionally, we observed a higher proportion of singletons SVs in coding or regulatory regions (Figure 2C), suggesting negative selection against deleterious SVs in functionally important regions of the genome. Overall, our results highlight the importance of evaluating rare SVs when studying genetic variation in human disease.

3.2 | SV quality evaluation and laboratory validation

Evaluation of the sensitivity of SV calling pipeline using synthetic mutations[47] (Section 2) revealed a sensitivity of 99.4% for 4000 deletions and 94.4% for 1500 inversions (Table S3). We did not perform an evaluation for insertions since the inserted sequences and positions are ambiguous in the simulation of synthetic mutations.

Then, we evaluated our SV call set against external SV databases. Approximately 50% of the high-quality SVs were detected in the Genome Aggregation Database (gnomAD, 292,307 SV sites), but there was less overlap with SVs in the 1000 Genomes Project (1 kg, 66,505 SV sites) (Figure S8). The difference was due to fewer samples in 1KG compared to gnomAD. The SV callset before high-quality filtering had

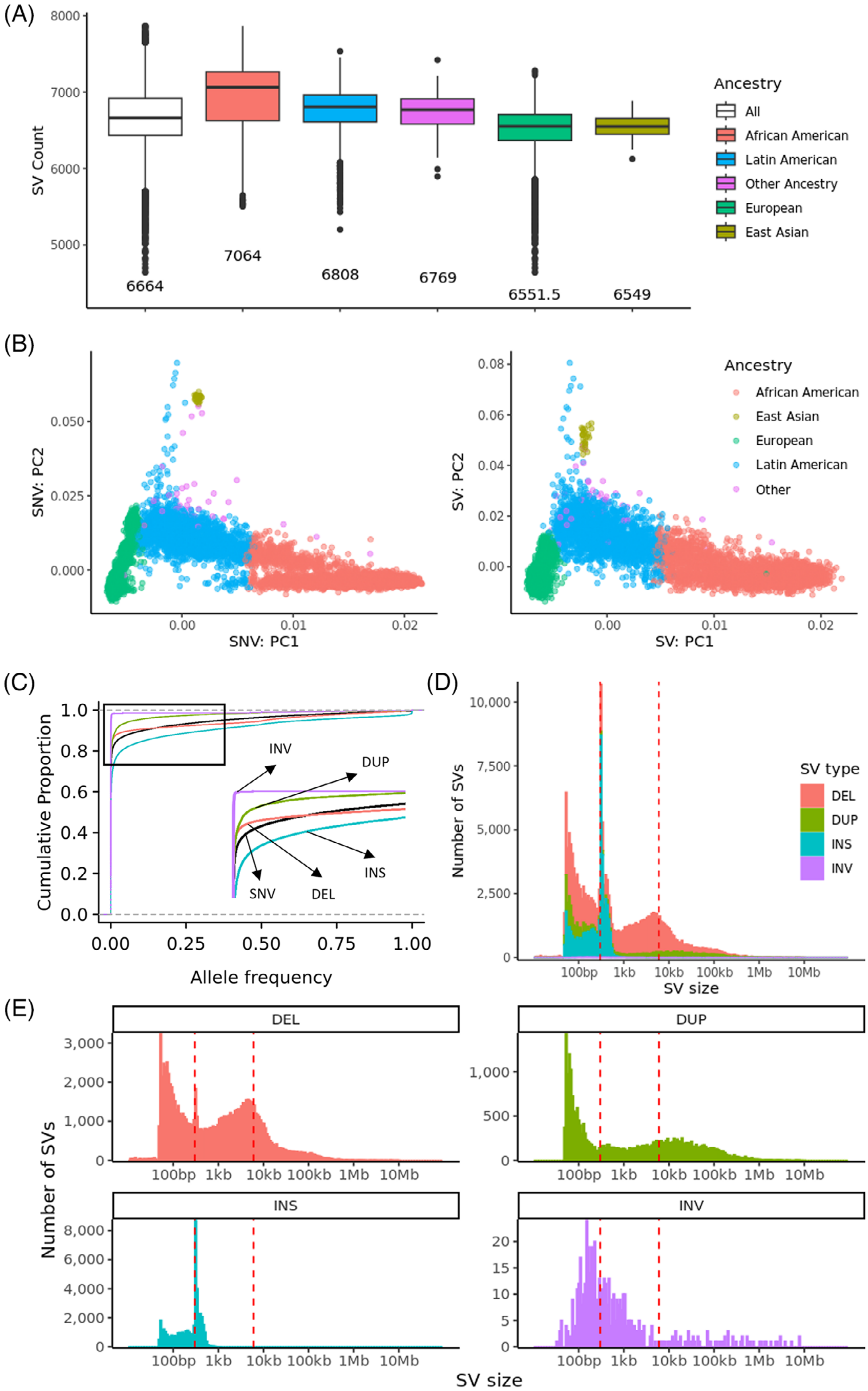

FIGURE 1 Characteristics of high-quality SVs. (A) Number of high-quality SVs per individual by ancestry. (B) Principal component analysis of high-quality SV with MAF > 0.01 and HWE > 1e-5. (C) The cumulative fractions of variants by AF. (D) The size distribution of high-quality SVs. (E) The size distribution of high-quality SVs by SV type. AF, allele frequency; HWE, Hardy–Weinberg equilibrium; MAF, minor allele frequency; SVs, structural variants.

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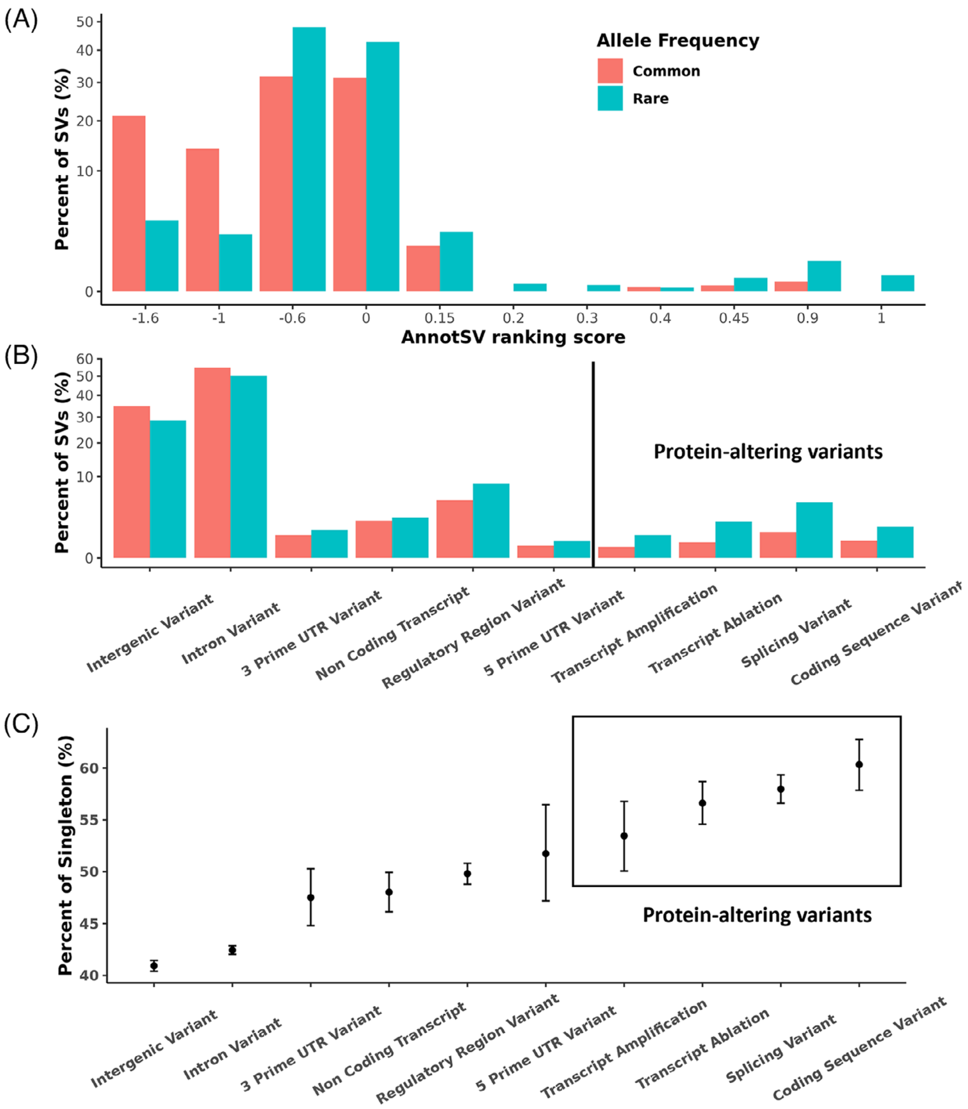

FIGURE 2 Functional annotation of SVs. (A) AnnotSV ranking scores of common (AF ≥ 0.01) and rare (AF < 0.01) high-quality SVs. The rare SVs are more likely to be deleterious with higher AnnotSV ranking scores. (B) VEP annotation of common (AF ≥ 0.01) and rare (AF < 0.01) high-quality SVs. The protein-altering SVs tend to be rare. (C) Percent of singletons in a specified functional category by VEP. The coding sequence variants include stop gain, start/stop lost, frameshift, inframe deletion/insertion, and missense mutations. SVs that were annotated (by VEP) to be able to cause transcript ablation/amplification, stop gain, start/stop lost, frameshift, inframe deletion/insertion, missense mutation, and affecting splice acceptor/donor were classified as protein-altering variants. AF, allele frequency; SVs, structural variants; VEP, Variant Effect Predictor.

a higher recall (a higher percentage of SVs from gnomAD and 1KG) at the cost of lower precision (a lower percentage of SVs confirmed by gnomAD and 1 kg) (Figures S8, S9).

We selected 95 SVs (534 genotype calls for 276 samples) for experimental validation (Table S4; Section 2). On the individual genotype level, an accuracy of 89% was achieved based on 276 validated samples for 95 SVs (Table S5). The genotype accuracy increased from 80% for SVs categorized as non-high-quality to 92% for those identified as high-quality SVs (Table S5). At the SV site level, we excluded two SVs that had only reference genotype calls (i.e., 0/0) among the validated samples. An SV site was considered validated only if all genotype calls for that SV were confirmed through PCR experiments (Section 2). Of the remaining 93 SVs, 77 were validated, resulting in a precision of 83%. When focusing on high-quality calls, the precision improves to 86%, with 61 out of 71 SVs being validated.

3.3 | SVs in LD with known AD risk loci

Previous GWAS identified AD risk loci were based on SNVs and small insertions/deletions.[10–12] However, SVs present larger genomic perturbations than SNVs and may have potential severe functional impacts. Therefore, it would be advantageous to consider SVs in LD with known AD risk loci as potential candidate causal variants. We identified 21 SVs (12 deletions, two duplications, and seven insertions) that are in LD with established AD GWAS loci[10–12] (Table 2, Figure S10). In particular, three deletions near or in *NCK2*, *NBEAL1*, and *TMEM106B* showed high LD ($R^2 > 0.9$) with GWAS signals. A 5.5 Kb deletion (chr2:105731359-105736864) located 8 Kb upstream of *NCK2* is in perfect LD ($R^2 = 0.99$) with rs143080277[12] (chr2:105749599, NCK2 intron, AF = 0.0046) and in moderate LD ($R^2 = 0.65$) with rs115186657[11] (chr2:105618971, ENSG00000287308 intron, AF = 0.0035). A 5.2 Kb deletion (chr2:203034369-203039560) in *NBEAL1* intron is in high LD ($R^2 = 0.94$) with rs139643391 (chr2:202878716),[12] which is a 3′ untranslated region (UTR) variant of *WDR12*. Though not in coding region, both deletions overlap with cCREs from ENCODE[50] and transcription factor binding sites (TFBSs) from ReMap[51] (Table 2). As for the 323 bp (chr7:12242077-12242399) Alu deletion located on the exon 8 of *TMEM106B*, it is in LD with *TMEM106B* intronic variants, rs5011436 (chr7:12229132, $R^2 = 0.92$)[11] and rs13237518 (chr7:12229967, $R^2 = 0.90$),[12] which are not only associated with the risk of AD but also protect carriers of *C9ORF72* repeat expansion from the risk of frontotemporal dementia.[63] Chemparathy et al.[64] also identified a deletion (chr7:12242077-12242399) and suggested that it is a leading candidate causal variant at *TMEM106B* reducing risk for frontotemporal lobar dementia with TDP pathology.

Other deletions that are in moderate LD ($0.2 < R^2 < 0.9$) with GWAS signals can impact exons, enhancers, transposons, and conserved regions (Table 2). A 365 bp deletion (chr7:28174681-28175045) in *JAZF1* intron overlapping evolutionally conserved sequence defined by phastCons and phyloP[65] is in LD ($R^2 = 0.41$) with rs1160871 (chr7:28129126).[12] A 446 bp deletion (chr10:122457302-122457747) extending into exon 2 of *ARMS2* is in LD ($R^2 = 0.24$) with rs7908662 (chr10:122413396, *PLEKHA1* intronic variant).[12] A 310 bp Alu deletion (chr12:113245316-113245625) in *TPCN1* intron is in LD ($R^2 = 0.73$) with rs6489896 (chr12:113281983).[12] A 310 bp deletion (chr14:106774952-106775261) in the IGH gene cluster and is in LD ($R^2 = 0.26$) with rs10131280 (chr14:106665591).[12] From functional annotations, the two deletions (chr12:113245316-113245625 and chr14:106774952-106775261) are possible regulatory elements as they are near cCREs and display a high density of TFBSs (Table 2).

The *MAPT* H1/H2 haplotype, defined by a 900 kb inversion and tagged by numerous SNVs, has been associated with several neurodegenerative diseases, including progress supranuclear palsy, frontotemporal disorders, Parkinson's disease, and AD.[27,28,66] We identified five deletions and two duplications in moderate LD ($R^2 = 0.35–0.67$, Table 2) with an H1/H2 tagging SNV (rs199515, chr17:46779275), which is associated with AD.[12] These SVs further confirmed the complex genomic structure in the region and highlight the difficulty in identifying the causal variants within the H1/H2 haplotype. Table 2 also describes seven high-quality insertions, excluding those in problematic regions (Section 2), that are in LD with AD GWAS signals. It should be noted that neither of the SNVs nor SVs showed genome-wide significance in association with AD using the current sample set (Table 2). This could be due to the fact that those variants have small effect sizes (such as variants in *TMEM106B*) or rather rare (such as variants in *NCK2*) (Table 2), thus necessitating a larger sample size for enough statistical power.

3.4 | SVs on AD risk/protective genes

We first focused on 10 SVs that were reported to be associated with AD in previous studies.[67–71] We identified SVs in our call set overlapping the 10 rare SVs (Table S9). A 417 Kb duplication (Figure S11) covering the *APP* is identified in one individual with early onset of AD at his age of 52. Subsequently, we noticed two other carriers of duplication who were dropped from the initial analysis due to failed quality control. One individual having the duplication was his sibling and developed AD at age of 49, and the other individual is his sibling's offspring who developed AD at age of 53. This finding provides compelling evidence that the duplication of *APP* is a rare cause of autosomal dominant early-onset AD.[18–20,22] A 7.68 Mb inversion covering the entire 21q21.2 is identified in one individual with early onset of AD at her age of 60 years. The inversion was experimentally validated, and the alignments showed clear breakpoints of the inversion (Figure S12). SVs spanning *EPHA5*, *HLA-DRA*, *FPR1*, and *DOPEY2* were previously reported as case-only variants. In our analysis, these SVs were also detected in non-AD samples. Notably, SVs in *EPHA5* and *HLA-DRA* are present in AD cases with an age of onset below 65. The 5.6 Kb deletion, covering exons 2–5 of *HLA-DRA* found in nine AD cases by Swaminathan et al.,[67] are present in eight samples in our analysis, including five AD cases (three showed early onset of AD with age < 65) and three unclear-AD-status individuals (two are diagnosed as progressive supranuclear palsy, and the remaining one is BRAAK stage 2). The 545 Kb duplication spanning *EPHA5*, previously reported by Hooli et al.[68] in two early-onset AD cases, was identified in four samples in our analysis. This includes two AD cases (one with early-onset AD under age 65), one case of progressive supranuclear palsy, and one control.

SVs on AD risk/protective genes could interfere with protein function and lead to disease. Therefore, we identified 77 high-confident SVs (Section 2), including 44 deletions, 15 duplications, and 18 insertions on 20 AD risk/protective genes determined by the ADSP gene verification committee (https://adsp.niagads.org/gvc-top-hits-list/). Nine deletions, five duplications, and 11 insertions have an allele count ≥ 5 (AF ranging from 0.0002 to 0.4690) (Table S6 and Table S7). Among these, only one insertion (chr14:73189448) within *PSEN1* intron showed a nominal significant association with AD (OR = 1.47, 95% CI 1.01–2.14, $p = 0.047$). None of these SVs were tagging known AD-associated SNVs. The remaining 35 deletions, 10 duplications, and seven insertions are ultra-rare (MAC < 5), of which 25 deletions, nine duplications, and one insertion are singletons (Table 3, Table S7, Figures S13–S17). Instead of calculating individual *p*-values given the limited statistical power due to low allele count, we performed an aggregated analysis of 35 ultra-rare deletions, 10 ultra-rare duplications, and seven insertions using SKAT-O test.[52] Overall, these ultra-rare variants were significantly associated with AD status ($p = 0.0050$). When analyzed by SV type, the SKAT-O test was significant ($p = 0.01$) for deletions but not for duplications ($p = 0.33$) and insertions ($p = 0.70$).

Notably, 14 of the 35 ultra-rare deletions and 8 of the 10 ultra-rare duplications are protein altering variants. For instance, we identified in

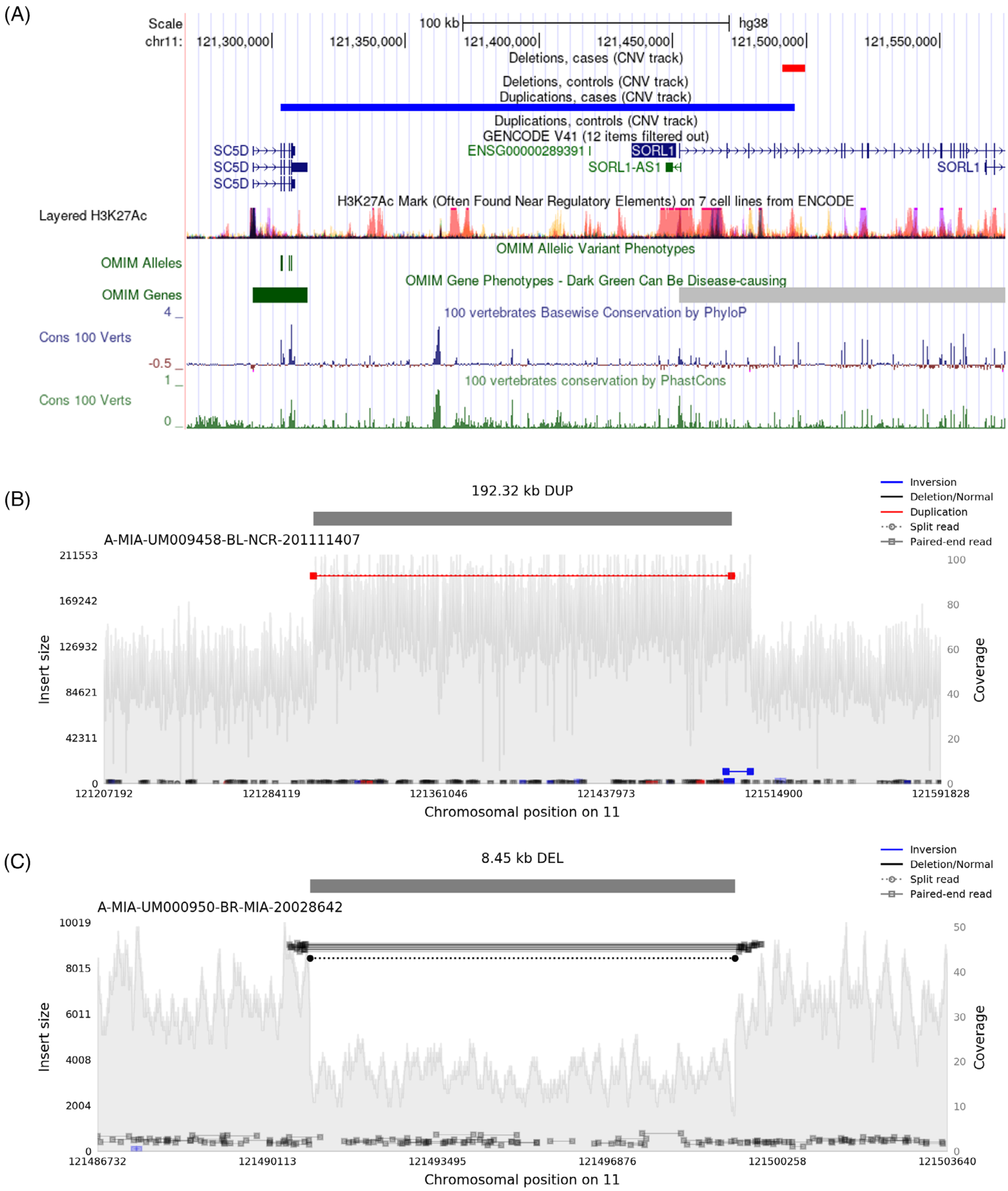

FIGURE 3 Ultra-rare deletion and duplication on SORL1. (A) Deletion and duplication of *SORL1*. (B) The 192 Kb duplication covers part of *SORL1* and *SC5D*. (C) The 8.45 Kb deletion covers exon 6 of *SORL1*.

SORL1 a 192 Kb duplication spanning exons 1–5 and an 8 Kb deletion affecting exon 6 (Figure 3). Previous studies indicated that *SORL1* deficiency can lead to AD through defects in the endolysosome-autophagy network,[72,73] and nearly all individuals with damaging SNVs in *SORL1* developed AD.[74] Eight out of nine individuals with *ABCA7* exonic deletions or duplications in our data (Figure S13) developed AD, supporting previous studies that observed loss-of-function *ABCA7* variants among AD cases.[75] We also found protein-altering ultra-rare deletions and duplications in *APP*, *PLCG2*, *PILRA*, *CASP7*, *MS4A6A*, *RIN3*, *APOE*, and *PSEN1* (Table 3). In particular, 17 of 21 individuals with ultra-rare deletions in *PLCG2* were AD cases (SKAT-O $p = 0.029$). Besides SVs on 20 AD genes determined by the ADSP gene verification committee, we also provided a full list of SVs on 78 other possible AD genes identified from GWAS[10–12] in Table S8.

3.5 | SV burden in AD

We performed burden tests of SVs, including deletions and duplications, insertions, and inversions separately and collectively, and found a moderate burden of deletions in AD cases (OR = 1.08, $p = 0.0064$), but no significant burden of duplications, insertions and inversions was detected (Table S10). The increased deletion burden in AD cases was particularly pronounced for singletons (OR = 1.07, $p = 0.0017$) and homozygous deletions (OR = 1.14, $p < 0.0001$). This is consistent with the burden of ultra-rare deletions in AD genes, in which 25 out of 35 ultra-rare deletions are singletons. The result suggests that singletons and homozygous deletions, which were not considered in previous association analyses, may be important contributors to the genetic basis of AD.

3.6 | SVs associated with AD and AD endophenotypes

In the discovery phase, we performed GWAS using 12,908 subjects (6328 AD cases and 6580 controls, excluding subjects with unknown AD diagnosis and SV quality outliers, Section 2), 10 common and 17 rare SVs were found associated with AD at a FDR < 0.2 (Table S11, Figure 4A, Figure S18). SVs near or on *RNA5SP293*, *RABGAP1*, *LMNTD1*, *LHFPL6*, *ADD3*, *ITPR2*, and *CLIC4* were confirmed by PCR validation. In replication phase (N = 15,528, 5051 AD cases and 10,477 controls, Table S1), the deletion (chr9:107896000-107900424) located near pseudogene *RNA5SP293* was replicated ($p < 0.05$, Table S11). The deletion displayed an odds ratio of 1.99 (95% CI 1.46–2.72, $p = 1.3 \times 10^{-5}$, Table S11) in all discovery and replication samples combined (N = 28,436, 11,379 AD cases and 17,057 controls). When examining 40 SNVs in LD ($R^2 > 0.2$) with the deletion, 29 SNVs showed nominal significance ($p < 0.05$) when all discovery and replication samples were combined (Table S12). The deletion contained a distal enhancer-like signature (chr9:107898669-107899003, ENCODE SCREEN ID: EH38E2715773[76]) which was evolutionally conserved as demonstrated by the high phastCons score and phyloP score[65] (Figure S19), and was predicted by a machine-learning model[77] to affect *KLF4* within the topological associating domain.

In addition, one duplication (chr9:122989084-122989182) on *RABGAP1* with a validation p of 0.1 is of potential interest. It displayed an odds ratio of 0.62 (95% CI 0.50–0.77, $p = 1.9 \times 10^{-5}$, Table S11) when all samples were combined. When examining 10 SNVs in LD ($R^2 > 0.2$) with the duplication, all showed nominal significance ($p < 0.05$) when all discovery and replication samples were combined (Table S12). From Chip-Seq data for various cell lines in ReMap,[51] the duplication (chr9:122989084-122989182) contains binding sites for C/EBP family of transcription factors (Figure S19), which have been reported to be a key transcription factor for *APOE* and preferentially mediates *APOE4* expression.[78] Particularly, compared to control noncarriers (N = 13), controls carrying the duplication (N = 585) showed significantly lower CSF Aβ levels ($p = 0.011$, Figure 4B). This conforms to the protective effect of the duplication as control carriers can bear a higher amyloid burden in the brain without developing dementia. On the other hand, compared to AD noncarriers (N = 4543), AD patients carrying the duplication (N = 45) showed significantly lower memory score ($p = 0.014$, Figure 4B), indicating the loss of protective effects in cases and faster decline once duplication carriers developed dementia.

Under a nominal $p < 0.05$, there are 2234 high-quality SVs not in the problematic regions (Section 2). Enrichment analysis of the 2234 SVs revealed an over-representation of neuronal function-related terms, such as axon development and synaptic membrane (Figure 4C). Among the 2234 SVs, 49 are protein-altering variants (Table S13), including protein-altering variants in genes that have been found to be related to AD, for example, *NTN3* and *CIB2*.[79,80]

Since a significant homozygous deletion burden is detected, we performed association using a recessive model, of which assumes that two copies of the alternative allele are required to alter the risk. As a result, seven SVs showed an FDR < 0.2 using the recessive model while none of them survived replication analysis (Table S11). In addition, we extended our association analysis to endophenotypes. No significant genomic inflation was observed for all endophenotypes (Figure S20), indicating that confounding factors are well adjusted. For endophenotypes, there are six common and six rare SVs with an FDR < 0.2 in the discovery phase while do not survive replication (Table S14, Figure S21).

4 | DISCUSSION

The complexity of generating high-quality SVs on WGS for SV association analysis is challenging, and a major concern is to ensure the analysis is not based on false positive SVs. To achieve this, we developed a pipeline to filter SVs and employed stringent criteria during the burden analysis to only include high-quality SVs. For each significant SV, we examined read coverages and other alignment signals by Samplot and performed experimental validations if samples were available in the lab (Section 2). Despite our efforts, false positive/negative calls on individual samples can still occur, which may undermine the result of the analysis. Therefore, we suggest a broader validation of significant SVs using long reads as the cost and accuracy of long reads improve rapidly.

We reported SVs in LD with known AD risk loci (such as SNVs in *NCK2*, *WDR12*, and *TMEM106B*) and on AD risk/protective genes (such as *APP*, *SORL1*, and *ABCA7*). Other than that, researchers can use our SV calling set to explore SVs on a particular gene of interest. For example, there are SVs on genes that might be related to the risk of disease by interacting with well-known AD genes (such as *PSEN2* and *APOE*). A deletion (chr1:226827423-226834076, near *PSEN2*) spanning the entire *lnc-PSEN2-7* and overlapping with a possible enhancer supported by H3K27ac signals (Figure S22) was identified in an individual (Latin American ancestry, inferred by GRAF-pop[32]), who had onset

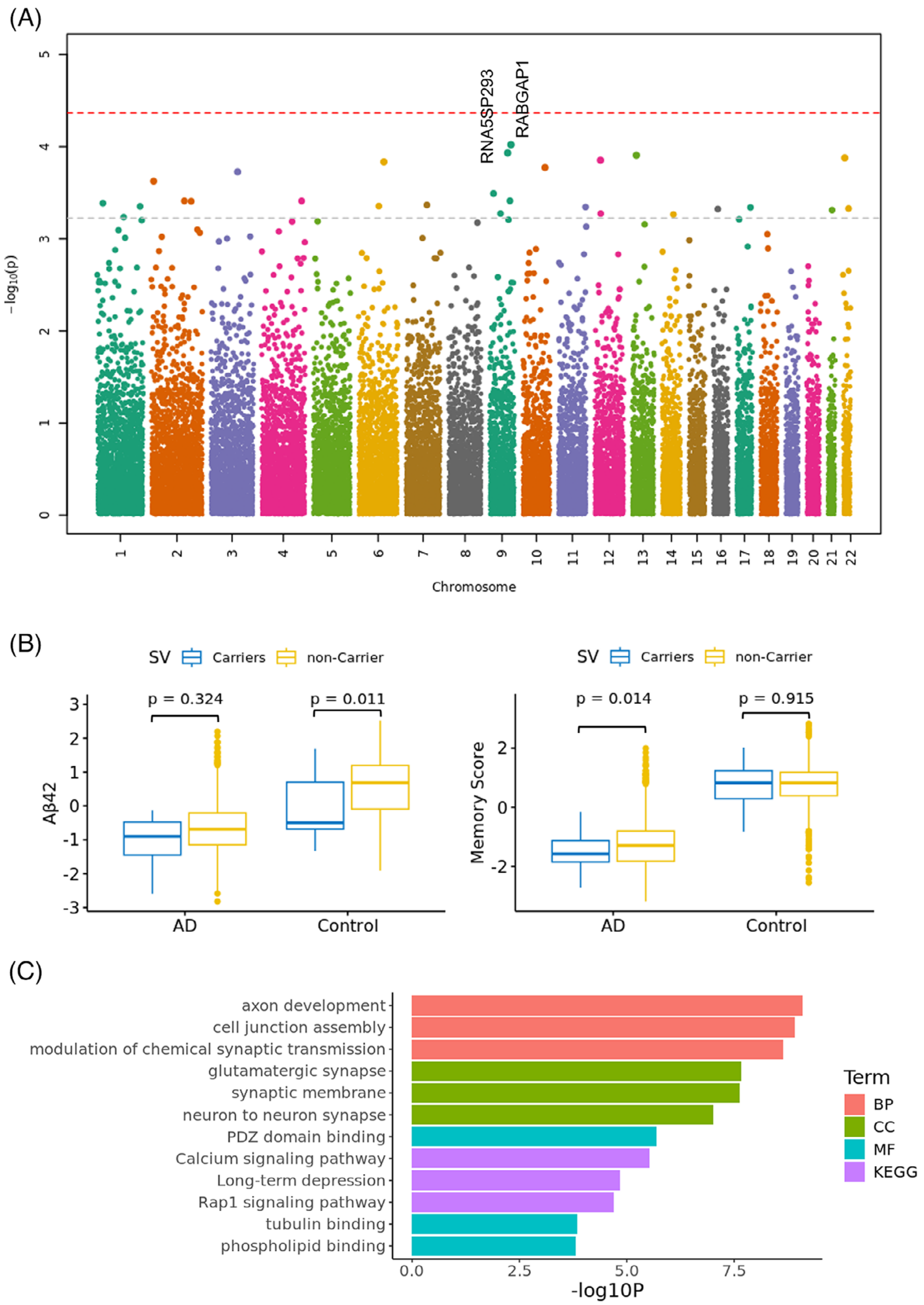

FIGURE 4 Association of SVs with AD and enrichment analysis. (A) Association of SVs with AD. The red line represents an FDR of 0.05. The gray line represents an FDR of 0.2. One SV near *RNA5SP293* and one SV on *RABGAP1* with a FDR < 0.2 and replication $p < 0.1$ were marked red. (B) Controls carrying the duplication on *RABGAP1* (chr9:122989084-122989182) showed significantly lower cerebrospinal fluid amyloid beta levels. AD patients carrying the duplication on *RABGAP1* (chr9:122989084-122989182) showed significantly lower memory scores. *p*-values were obtained by the Wilcoxon rank-sum test. (C) Enrichment analysis for high-quality SVs (nominal $p < 0.05$) that are not in problematic regions. AD, Alzheimer's disease; BP, biological process; CC, cellular component; FDR, false discovery rate; MF, molecular function; KEGG, Kyoto Encyclopedia of Genes and Genomes; SVs, structural variants.

of AD symptoms at age 71 years old. We also observed in one AD case an exonic deletion in *MPO* (Figure S23), a gene that has been reported to affect AD risk through interact with *APOE*.[81]

Overall, we demonstrated the role of SVs in AD by revealing a significant burden of deletions (particularly, homozygous and singleton deletions) in AD. However, no significant burden was observed for insertions, duplications, and inversions. It is important to acknowledge the presence of undetected SVs in our analysis. Each human genome is estimated to contain > 20,000 SVs, many of which are located in regions where short reads are unmappable.[82] Particularly, current methods may lack the power to identify inversion variations, which are primarily composed by complex SVs rather than canonical inversions.[45] Therefore, it should be recognized that not all classes of SVs have been completely resolved while reporting the burden of SVs. Moreover, only four singletons were included among 95 SVs that went through PCR validation due to the limitation in sample DNA availability. Though we did visual inspection by read coverages and other alignment signals using Samplot, validation of more singleton SVs is needed to ensure the reported singleton burden in AD is accurate and robust.

Following burden analysis, a few common and rare variants were discovered by association analysis. Based on the analysis of the independent replication dataset, a deletion (chr9:107896000-107900424) located 18 Kb away from pseudogene *RNA5SP293* and a duplication (chr9:122989084-122989182) on *RABGAP1* are prioritized over other significant SVs. Moreover, another of our work,[83] which imputed the observed SVs into a larger cohort (N = 38,271), showed two other rare SVs (chr6:113757605-INS and chr7:102726253-102733877-DEL) exhibited a validation p of 7.37×10^{-7} and 2.32×10^{-4}.

Given that age of onset is a discriminating factor in AD genetics, we conducted analyses of significant signals for late-onset cases ($\geq$ 65) versus controls, as well as early-onset cases (< 65) versus controls. Notably, our findings revealed that the signals predominantly stemmed from late-onset cases (Table S15), which represent the majority of AD cases in our dataset. The AF of SVs can vary across ancestries. For instance, the deletion at chr9:107896000-107900424 is rare in European populations (AF < 0.001) and Latin American populations (AF < 0.005) but more frequent in African American populations (AF = 0.01; Table S16). Conversely, the duplication at chr9:122989084-122989182 is more prevalent in European (AF > 0.005) and Latin American populations (AF > 0.003) compared to African American populations (AF $\leq$ 0.001; Table S16). Therefore, we also conducted association analyses stratified by ancestry, followed by a meta-analysis across ancestries, which yielded results consistent with those obtained from pooled samples.

Moreover, the identified common SVs from the discovery phase displayed moderate ORs (OR, 0.67–1.36) and were not significant in the replication phase, indicating that the power of association analysis for SVs is still constrained. This could be due to limitations in the selection of study subjects, availability of WGS data, and technical constraints of short reads. First, the population structure, age distribution, and the ratio of case/control are different in the discovery and replication phase: there were much fewer cases with European and African American ancestry and many more cases with Latin American ancestry in the replication phase compared to the discovery phase; the mean age of controls in the discovery phase is 71.27 compared to 77.40 in the replication phase; the number of controls is twice the number of cases in the replication phase. Second, our sample size was smaller compared to SNV GWAS by genotyping arrays due to the limited availability of WGS data. Finally, not all SVs can be discovered by short-read data, and even fewer high-quality ones underwent association analysis. Therefore, future SV GWASs are needed when considering SVs called from long reads data from a large-scale diverse sample set.

Our study provided a valuable resource for the analysis of SVs in AD. We identified SVs from WGS data across a large cohort of AD participants with diverse ancestry. We reported SVs tagging AD risk SNVs, providing new mechanisms of action for GWAS signals. Deleterious rare SVs on well-known AD genes have been discovered. We found a higher burden of ultra-rare SVs on AD genes, and overall, higher burden of homozygous and singleton deletions in AD patients. Finally, our association analysis revealed a few potential candidate SVs and genes that are worthy of further study.

ACKNOWLEDGMENTS

H.W. and P.L.C. report grant support from RF1-AG074328 and P30-AG072979. A.T., Y.Q.S., and J.Y.T. report grant support from RF1-AG074328. Y.Y.L. reports grant support from U54-AG052427 and U24-AG041689. L.S.W. reports grant support from U24-AG041689, U54-AG052427, U01-AG032984, U01-AG058654, and P30AG072979. L.A.F. reports grant support from U54-AG052427, U01-AG058654, U01-AG062602, R01-AG048927, and P30-AG072978. W.P.L. reports grant support from RF1-AG074328, P30-AG072979, U54-AG052427, and U24-AG041689.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. Author disclosures are available in Supporting Information.

CONSENT STATEMENT

All human subjects provided informed consent.

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SUPPORTING INFORMATION

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How to cite this article: Wang H, Dombroski BA, Cheng P-L, et al. Structural variation detection and association analysis of whole-genome-sequence data from 16,543 Alzheimer's disease sequencing project subjects. *Alzheimer's Dement*. 2025;21:e70277. https://doi.org/10.1002/alz.70277